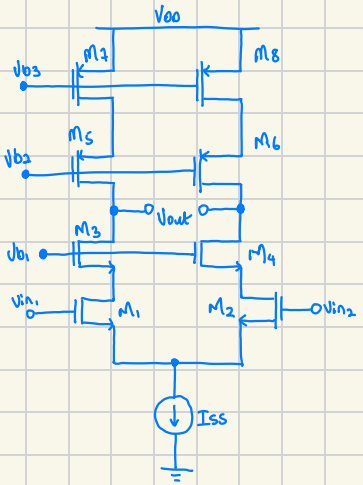
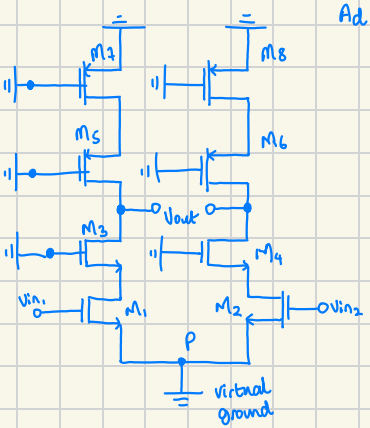
ELG 3136

Lecture 3

Differential Amplifier



AC analysis: Ground DC



$Vin_1 = Vin + Vn$

$Vin_2 = -Vin + Vn$

$Vout_1 = Vout + Vn$

$Vout_2 = -Vout + Vn$

$$Ad = \frac{Vout_2 - Vout_1}{Vin_2 - Vin_1} = \frac{-Vout + Vn - (Vout + Vn)}{-Vin + Vn - (Vin + Vn)}$$

$$= \frac{-Vout + \cancel{Vn} - Vout - \cancel{Vn}}{-Vin + \cancel{Vn} - Vin - \cancel{Vn}} = \frac{-2Vout}{2Vin} = Av.$$

Traditional Amplifier

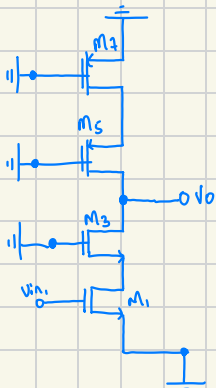
$$Av = \frac{Vout_1}{Vin_1} = \frac{Vout + Vn}{Vin + Vn}$$

Notice: 2 input voltages & output voltages taken from 2 voltages.

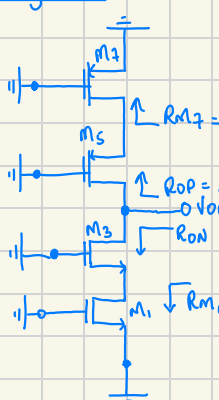
Using BRO CODE

$Gm = gm_1$

$Av = -Gm\,Rout$



Finding Rout : Disable all sources



$R_{M7} = r_{o7} = R_S$

$R_{OP} = g_{m5} r_{o5} r_{o7}$

$Vout \leftarrow Rout = R_{ON} \,//\, R_{OP} = (g_{m3} r_{o3} r_{o1} \,//\, g_{m5} r_{o5} r_{o7})$

$R_{ON} = g_{m3} r_{o3} r_{o1}$

$R_{M1} = r_{o1} = R_S$

$$Av = -g_{m1}(g_{m3} r_{o3} r_{o1} \,//\, g_{m5} r_{o5} r_{o7})$$